COMP3311 Week 2 Monday Lecture

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◆ Week 02

In today's lecture ...

- SQL Data Definition Language (DDL)
- Mapping ER → Relational SQL

Things to do ...

- Tutorials start this week
- Quiz due Friday midnight
- Assignment 1 out later in the week
- Set up your PostgreSQL server
  - 390/7 DB students have logged in to vlad62 and have .bashrcmyap
  - come to a Help Session if you're having trouble installing PostgreSQL

◆ Email and Context

When sending email to us (cs3311), please include your zID

Before typing commands, think about the context:

Prompt	Context
$	in Linux shell run Unix commands, e.g. **ls -ld**
db=#	**psql** run SQL commands, e.g. **select, update**
db>	**mysql(sql)** run SQL commands, e.g. **select, update**
none	**vim**, you are doomed but try **:q:q:** **ZZ**

◆ n:m:p Relationships

Two possible ER models for "prescribes" in the medical scenario

Could be done as a 3-way relationship, or using a new entity

◆ Data Models (recap)

Entity-relationship (ER) data model

- attributes, entities, relationships, subclasses
- relationship variations: total/partial, n:m, 1:1

Relational data model

- attributes, tuples, relations
- attribute = name + domain/type
- tuple = list of attributes, based on attribute definitions
- relation = set of tuples, based on a tuple definition

E.g. **Student(zID:integer, name:string, WAM:float, ...)**

◆ Mapping ER to SQL

Above example gives informal description of relational schema

Need a more formal way of describing relational schemas

SQL data definition language (DDL) provides this

SQL is an implementation of relational data model

- relations → tables, tuples → tuples, attributes → columns/fields

Mapping ER to SQL

- entity sets → tables, entities → rows/tuples, attributes → columns/fields
- relationships → tables or foreign keys
- multi-valued attributes/weak entities/subclasses → ?

◆ SQL Schemas (recap)

Primary SQL DDL construct is table creation:

```
create table TableName (
  attr1_name type [constraint],
  attr2_name type [constraint],
  attr3_name type [constraint],
  ...
  primary key (attr1_name ,
  foreign key (attr2_name
              references OtherTable (attr1_name )
)
```

SQL schema = collection of table definitions, including constraints

◆ SQL Types

Built-in types

- numbers: **integer**, **numeric(n)**, **real**
- strings: **char(n)**, **varchar(n)**, **text**
- time: **date**, **time**, **timestamp**, **interval**
- **boolean**, monetary, geometric, enumerated, ...

Make your own

```
create domain Name as Type Constraint;
create type Name as enum ('val1','val2',...);
```


◆ SQL Constraints

Constraints in SQL DDL

- on attributes e.g. **integer, check (x > 0), not null**
- on attributes e.g. **text** can use regular expressions
- on table e.g. **unique, primary key (a,b,c)**
- between tables e.g. **foreign key (a) references T(y)**

Tuples which do not satisfy constraints cannot be added to DB

Gives strong guarantee that the data is valid (internally consistent)

But does not guarantee that it reflects reality

◆ Exercise: Constraints

Constraint = SQL expression limiting possible values

Define type + constraints for

- positive integers
- marks (range 0..100)
- unsw course codes (e.g. COMP3311)
- unsw term codes (e.g. 2571)
- student IDs (e.g. z5432123)
- person's name (alpha + space + - + ')
- colours (e.g. red, green, blue)
- grades (e.g. HD, DN, CR, PS, FL, UF)

◆ Exercise: ER-to-SQL Mapping

Describe this ER model as an SQL schema

Use domain definitions to make types more precise.

Use integer **id** attributes for all entity tables.

◆ Exercise: ER-to-SQL Mappings

For each of the following ER-to-relational mappings:

- give an SQL rendering of the relational version
- choose obvious and simple domains for attributes

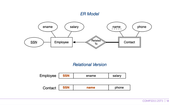
◆ Mapping Strong Entities

An entity set E with atomic attributes $a_1, a_2 ... a_n$

maps to

A relation R with attributes (columns) $a_1, a_2 ... a_n$

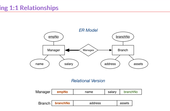
Example:



Note: the key is preserved in the mapping.

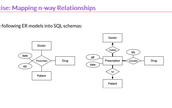
◆ Mapping Weak Entities

Example:



◆ Mapping N:M Relationships

Example:



◆ Mapping 1:N Relationships

Example:

◆ Mapping 1:1 Relationships

Example:

◆ Mapping 1:1 Relationships (cont)

If there is no reason to favour one side of the relationship ...

◆ Mapping n-way Relationships

Relationship mappings above assume binary relationship.

If multiple entities are involved:

- n:m generalises naturally to n:m:p
  - include foreign key for each participating entity
  - include any other attributes of the relationship
- other multiplicities (e.g. 1:n:m) ...
  - need to be mapped the same as n:m:p
  - so not quite an accurate mapping of the ER

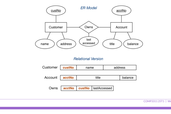
◆ Exercise: Mapping n-way Relationships

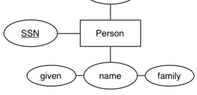
Convert the following ER models into SQL schemas:

◆ Mapping Composite Attributes

Composite attributes are mapped by concatenation or flattening.

Example:





Relational Version #2

Person

SSN	given	family	.....

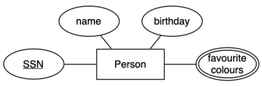
.....	number	street	suburb	pcode

❖ Mapping Multi-valued Attributes (MVAs)

MVAs are mapped by a new table linking values to their entity.

Example:

ER Model



Relational Version

Person	SSN	name	birthday
FavColour	SSN	colour	

❖ Mapping Subclasses

Three different approaches to mapping subclasses to tables:

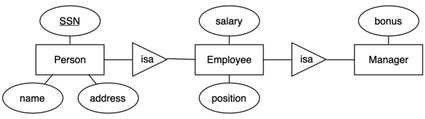
- ER style
  - each entity becomes a separate table,
  - containing attributes of subclass + FK to superclass table
- object-oriented
  - each entity becomes a separate table,
  - inheriting all attributes from all superclasses
- single table with nulls
  - whole class hierarchy becomes one table,
  - containing all attributes of all subclasses (null, if unused)

Which mapping is best depends on how data is to be used.

❖ Mapping Subclasses (cont)

Example of ER-style mapping:

ER Model



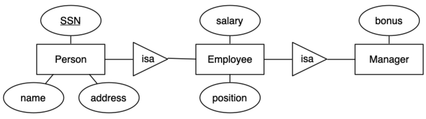
Relational Version

Person	SSN	name	address
Employee	SSN	salary	position
Manager	SSN	bonus	

❖ Mapping Subclasses (cont)

Example of object-oriented mapping:

ER Model



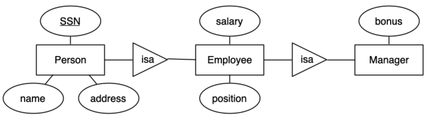
Relational Version

Person	SSN	name	address			
Employee	SSN	name	address	salary	position	
Manager	SSN	name	address	salary	position	bonus

❖ Mapping Subclasses (cont)

Example of single-table-with-nulls mapping:

ER Model

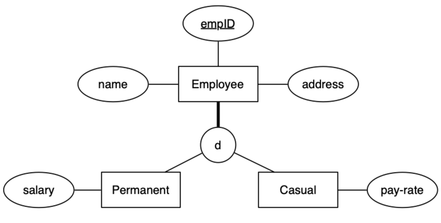


Relational Version

Person	SSN	name	address	salary	position	bonus

NULL for Person who is not Employee

NULL for Employee who is not Manager

❖ Exercise: Class Hierarchy ER-to-SQL

Convert the following class hierarchy to SQL using ER mapping: